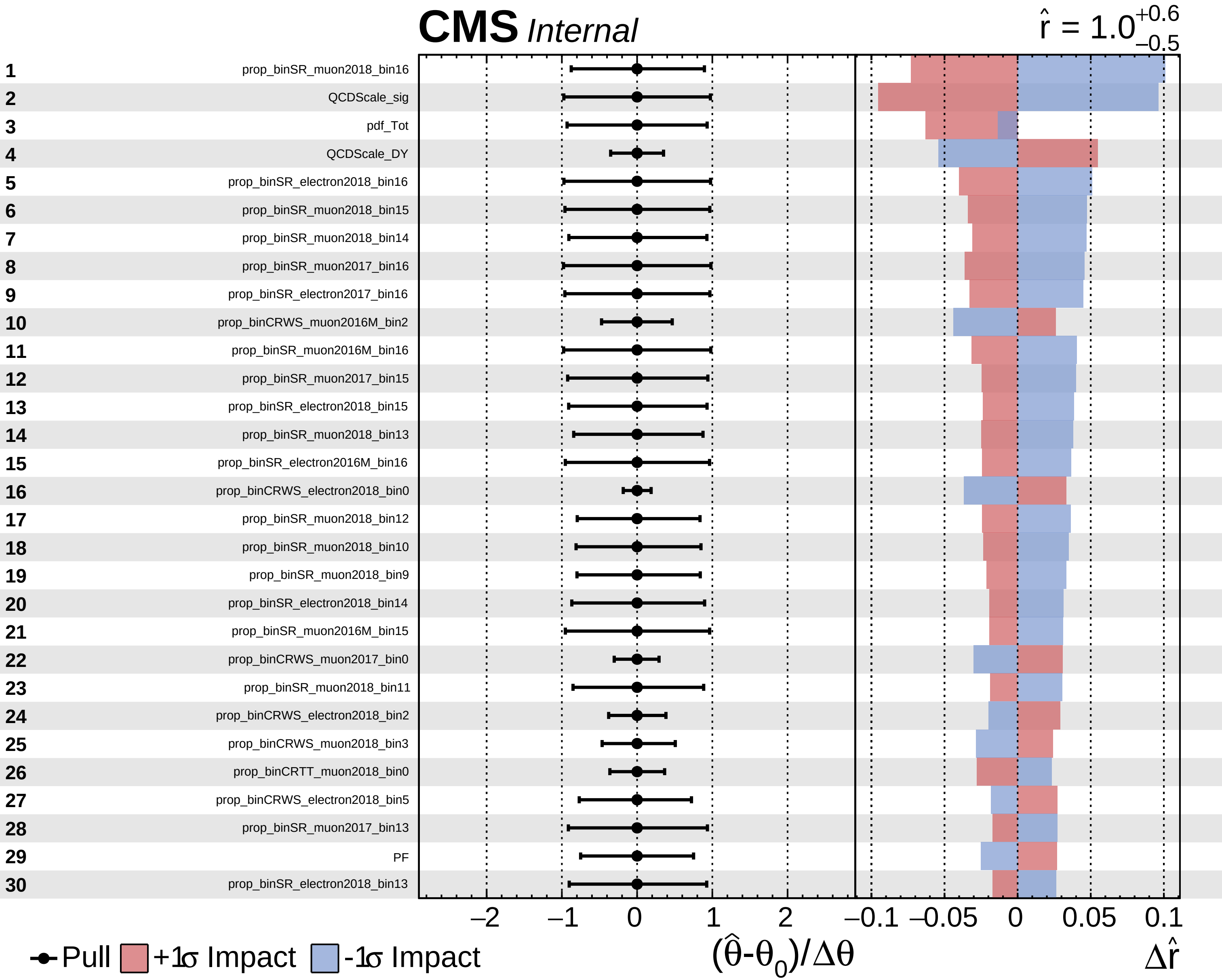
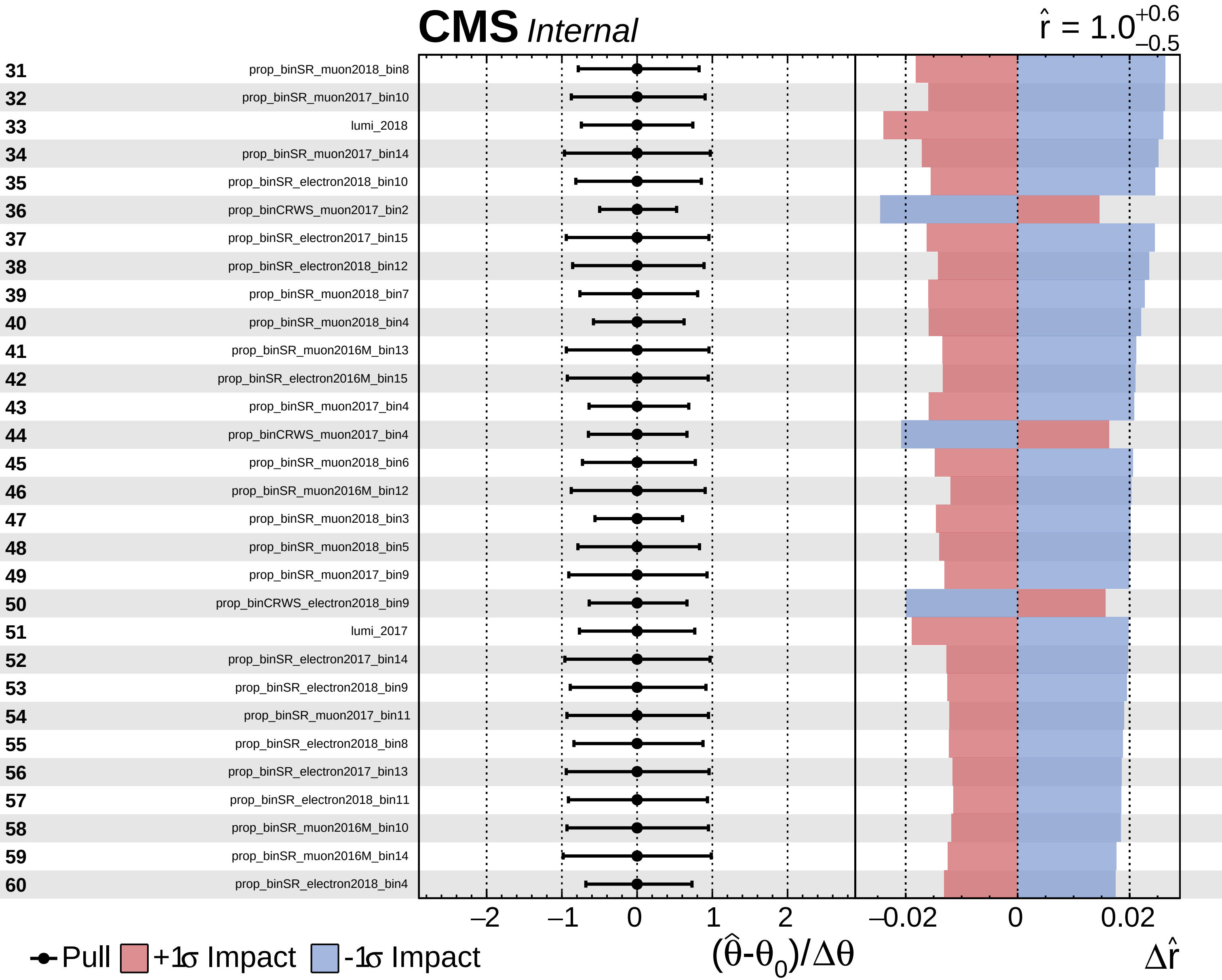
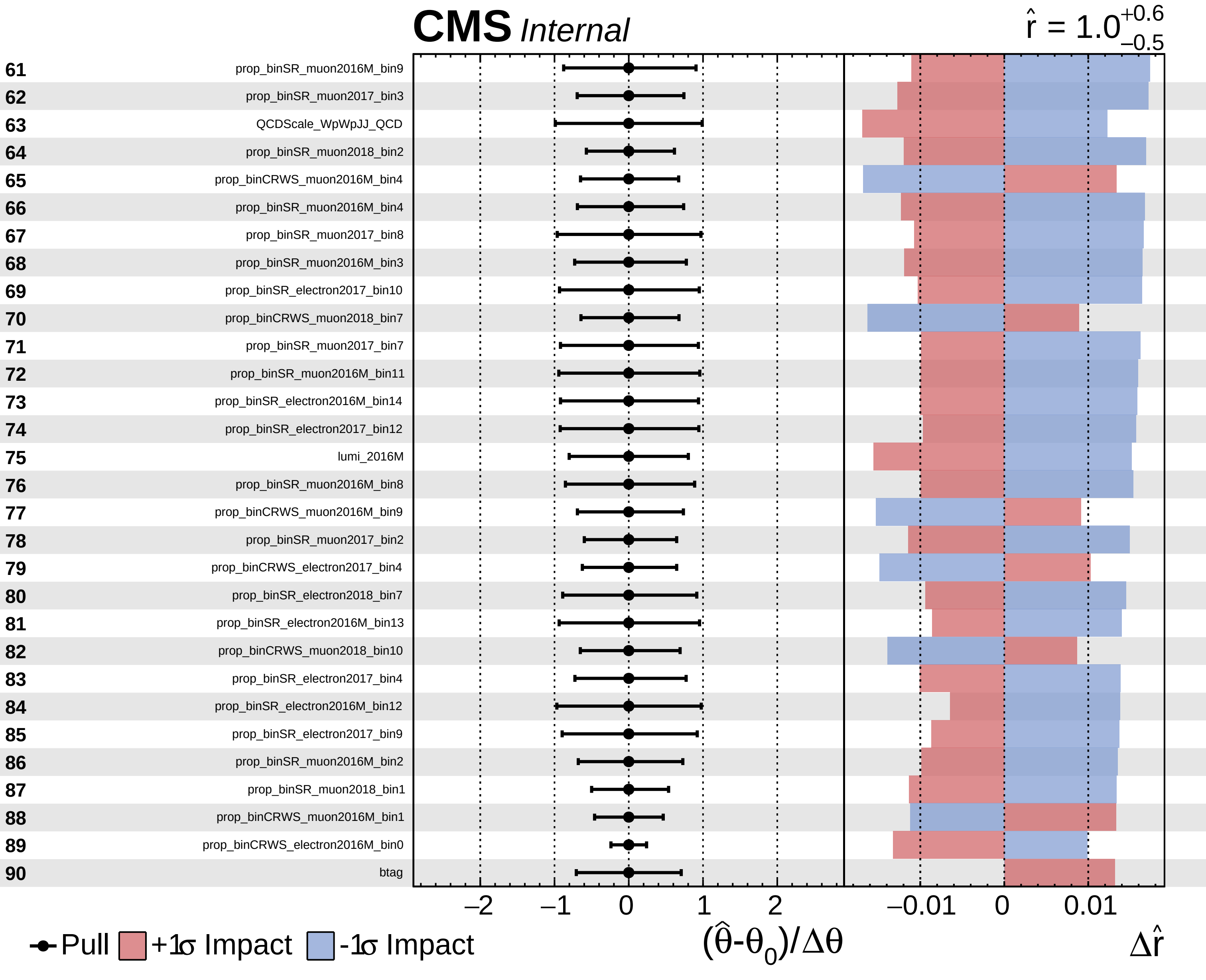
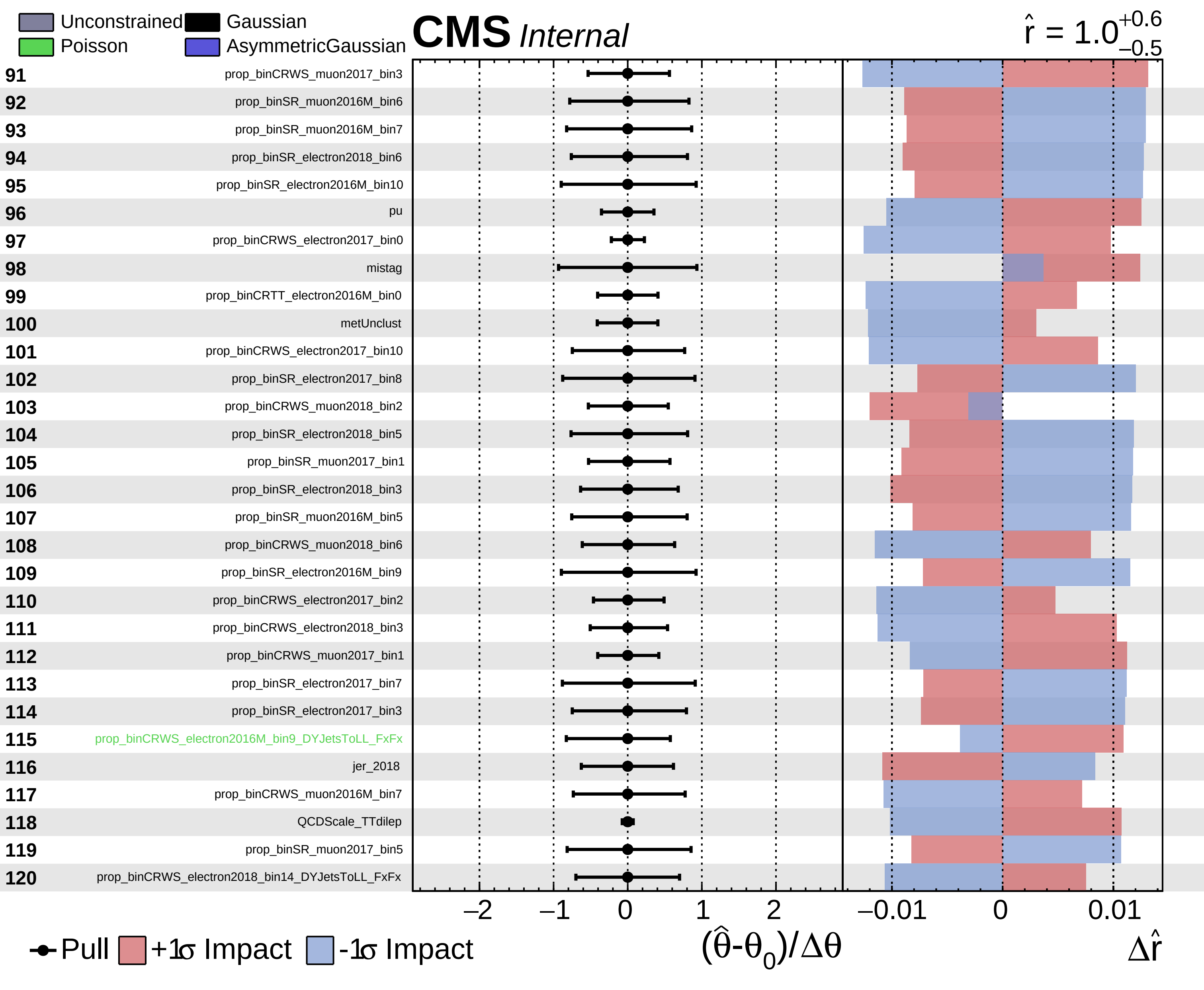


CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
1 prop_binSR_muon2018_bin16
2 QCDScale_sig
3 pdf_Tot
4 QCDScale_DY
5 prop_binSR_electron2018_bin16
6 prop_binSR_muon2018_bin15
7 prop_binSR_muon2018_bin14
8 prop_binSR_muon2017_bin16
9 prop_binSR_electron2017_bin16
10 prop_binCRWS_muon2016M_bin2
11 prop_binSR_muon2016M_bin16
12 prop_binSR_muon2017_bin15
13 prop_binSR_electron2018_bin15
14 prop_binSR_muon2018_bin13
15 prop_binSR_electron2016M_bin16
16 prop_binCRWS_electron2018_bin0
17 prop_binSR_muon2018_bin12
18 prop_binSR_muon2018_bin10
19 prop_binSR_muon2018_bin9
20 prop_binSR_electron2018_bin14
21 prop_binSR_muon2016M_bin15
22 prop_binCRWS_muon2017_bin0
23 prop_binSR_muon2018_bin11
24 prop_binCRWS_electron2018_bin2
25 prop_binCRWS_muon2018_bin3
26 prop_binCRTT_muon2018_bin0
27 prop_binCRWS_electron2018_bin5
28 prop_binSR_muon2017_bin13
29 PF
30 prop_binSR_electron2018_bin13
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

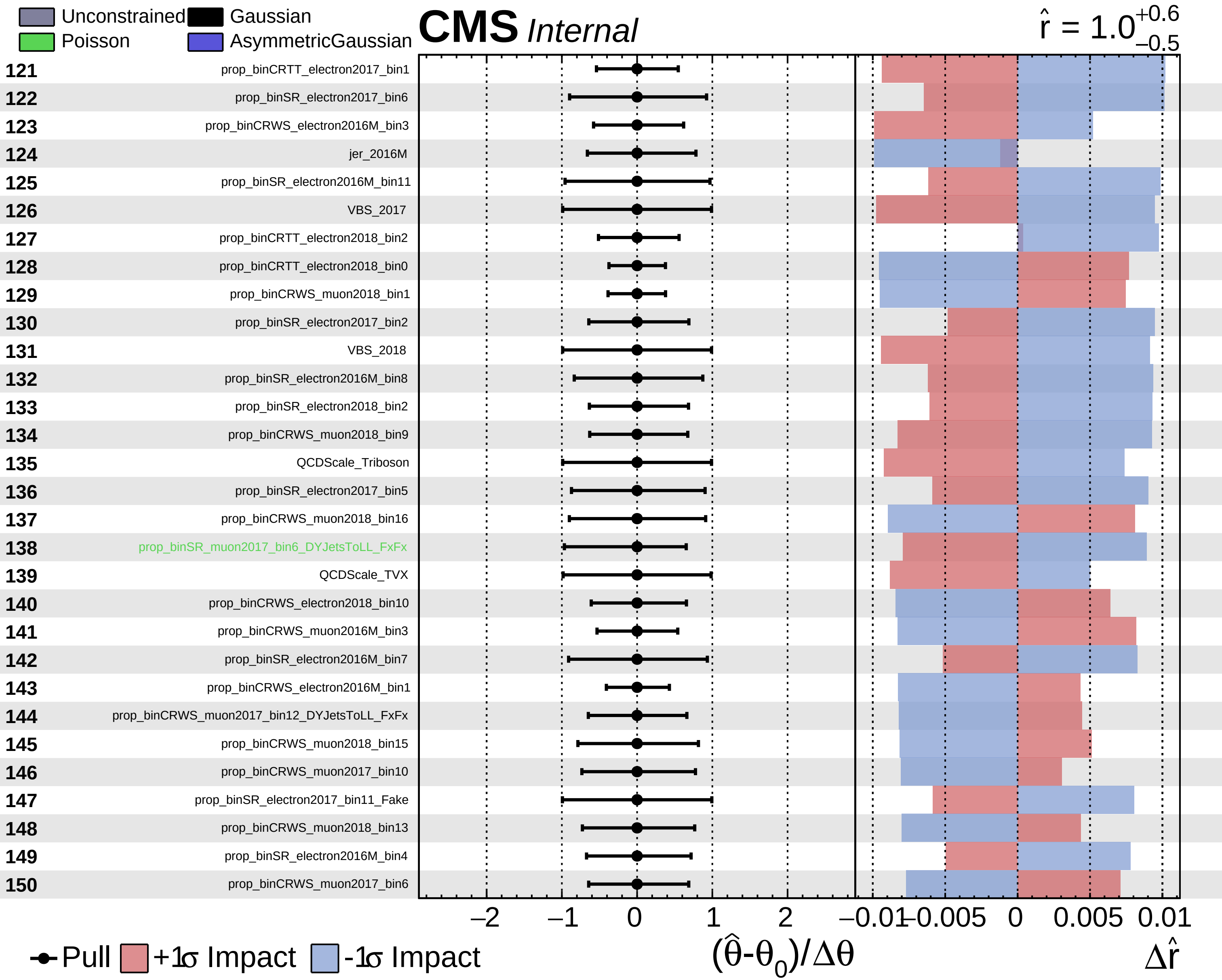


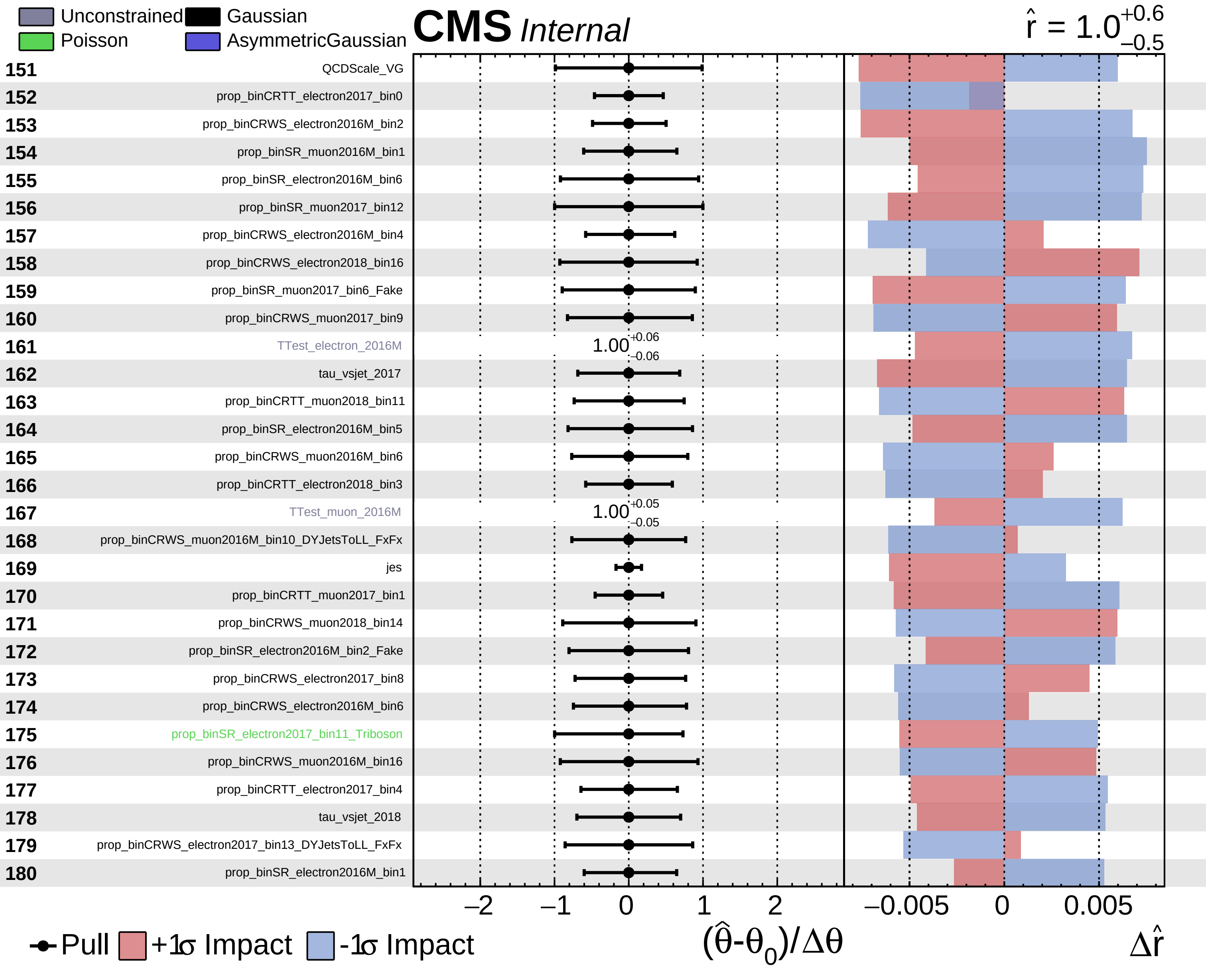
CMS Internal
r̂ = 1.0 +0.6 −0.5
31 prop_binSR_muon2018_bin8
32 prop_binSR_muon2017_bin10
33 lumi_2018
34 prop_binSR_muon2017_bin14
35 prop_binSR_electron2018_bin10
36 prop_binCRWS_muon2017_bin2
37 prop_binSR_electron2017_bin15
38 prop_binSR_electron2018_bin12
39 prop_binSR_muon2018_bin7
40 prop_binSR_muon2018_bin4
41 prop_binSR_muon2016M_bin13
42 prop_binSR_electron2016M_bin15
43 prop_binSR_muon2017_bin4
44 prop_binCRWS_muon2017_bin4
45 prop_binSR_muon2018_bin6
46 prop_binSR_muon2016M_bin12
47 prop_binSR_muon2018_bin3
48 prop_binSR_muon2018_bin5
49 prop_binSR_muon2017_bin9
50 prop_binCRWS_electron2018_bin9
51 lumi_2017
52 prop_binSR_electron2017_bin14
53 prop_binSR_electron2018_bin9
54 prop_binSR_muon2017_bin11
55 prop_binSR_electron2018_bin8
56 prop_binSR_electron2017_bin13
57 prop_binSR_electron2018_bin11
58 prop_binSR_muon2016M_bin10
59 prop_binSR_muon2016M_bin14
60 prop_binSR_electron2018_bin4
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
61 prop_binSR_muon2016M_bin9
62 prop_binSR_muon2017_bin3
63 QCDScale_WpWpJJ_QCD
64 prop_binSR_muon2018_bin2
65 prop_binCRWS_muon2016M_bin4
66 prop_binSR_muon2016M_bin4
67 prop_binSR_muon2017_bin8
68 prop_binSR_muon2016M_bin3
69 prop_binSR_electron2017_bin10
70 prop_binCRWS_muon2018_bin7
71 prop_binSR_muon2017_bin7
72 prop_binSR_muon2016M_bin11
73 prop_binSR_electron2016M_bin14
74 prop_binSR_electron2017_bin12
75 lumi_2016M
76 prop_binSR_muon2016M_bin8
77 prop_binCRWS_muon2016M_bin9
78 prop_binSR_muon2017_bin2
79 prop_binCRWS_electron2017_bin4
80 prop_binSR_electron2018_bin7
81 prop_binSR_electron2016M_bin13
82 prop_binCRWS_muon2018_bin10
83 prop_binSR_electron2017_bin4
84 prop_binSR_electron2016M_bin12
85 prop_binSR_electron2017_bin9
86 prop_binSR_muon2016M_bin2
87 prop_binSR_muon2018_bin1
88 prop_binCRWS_muon2016M_bin1
89 prop_binCRWS_electron2016M_bin0
90 btag
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

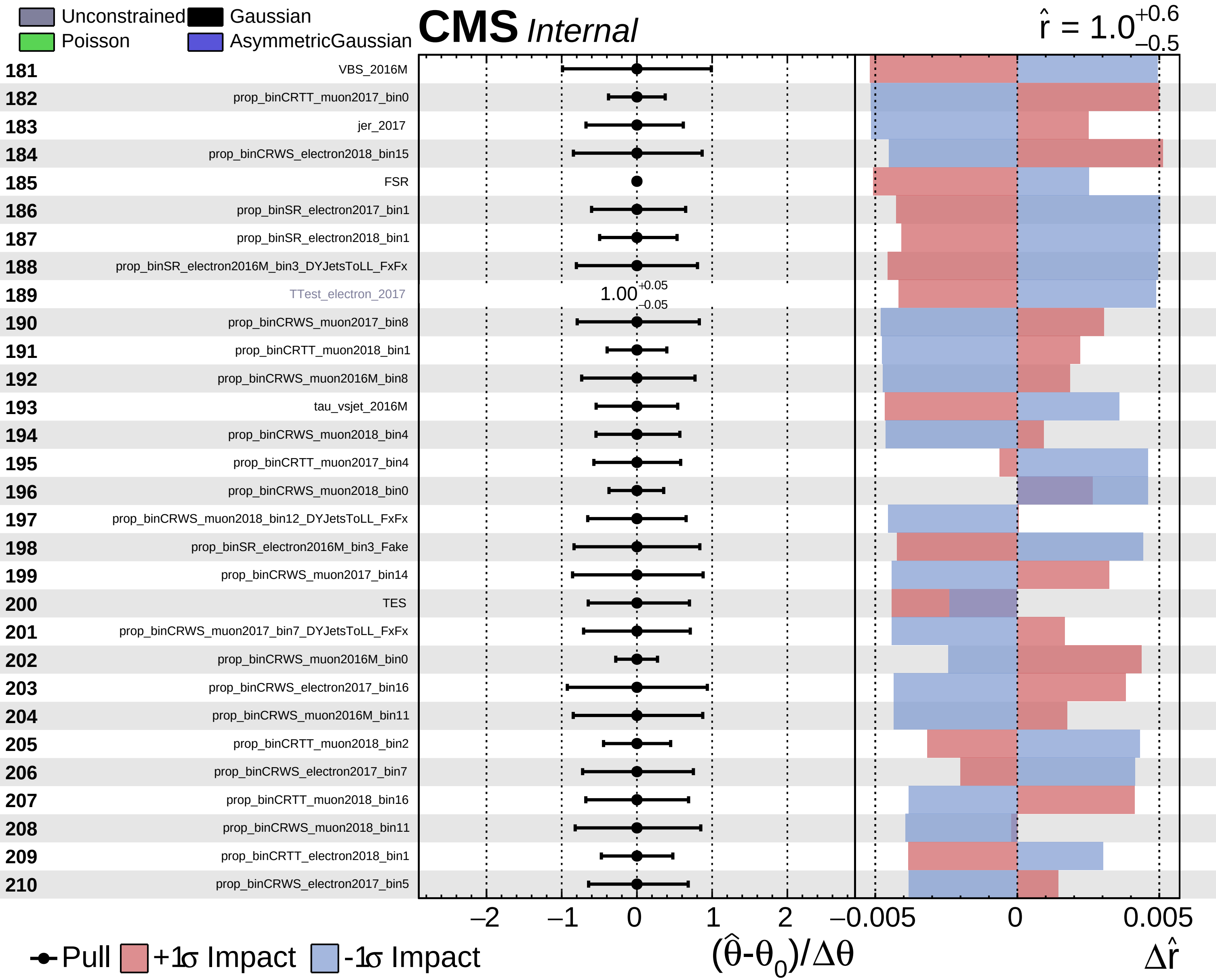


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
91 prop_binCRWS_muon2017_bin3
92 prop_binSR_muon2016M_bin6
93 prop_binSR_muon2016M_bin7
94 prop_binSR_electron2018_bin6
95 prop_binSR_electron2016M_bin10
96 pu
97 prop_binCRWS_electron2017_bin0
98 mistag
99 prop_binCRTT_electron2016M_bin0
100 metUnclust
101 prop_binCRWS_electron2017_bin10
102 prop_binSR_electron2017_bin8
103 prop_binCRWS_muon2018_bin2
104 prop_binSR_electron2018_bin5
105 prop_binSR_muon2017_bin1
106 prop_binSR_electron2018_bin3
107 prop_binSR_muon2016M_bin5
108 prop_binCRWS_muon2018_bin6
109 prop_binSR_electron2016M_bin9
110 prop_binCRWS_electron2017_bin2
111 prop_binCRWS_electron2018_bin3
112 prop_binCRWS_muon2017_bin1
113 prop_binSR_electron2017_bin7
114 prop_binSR_electron2017_bin3
115 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
116 jer_2018
117 prop_binCRWS_muon2016M_bin7
118 QCDScale_TTdilep
119 prop_binSR_muon2017_bin5
120 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

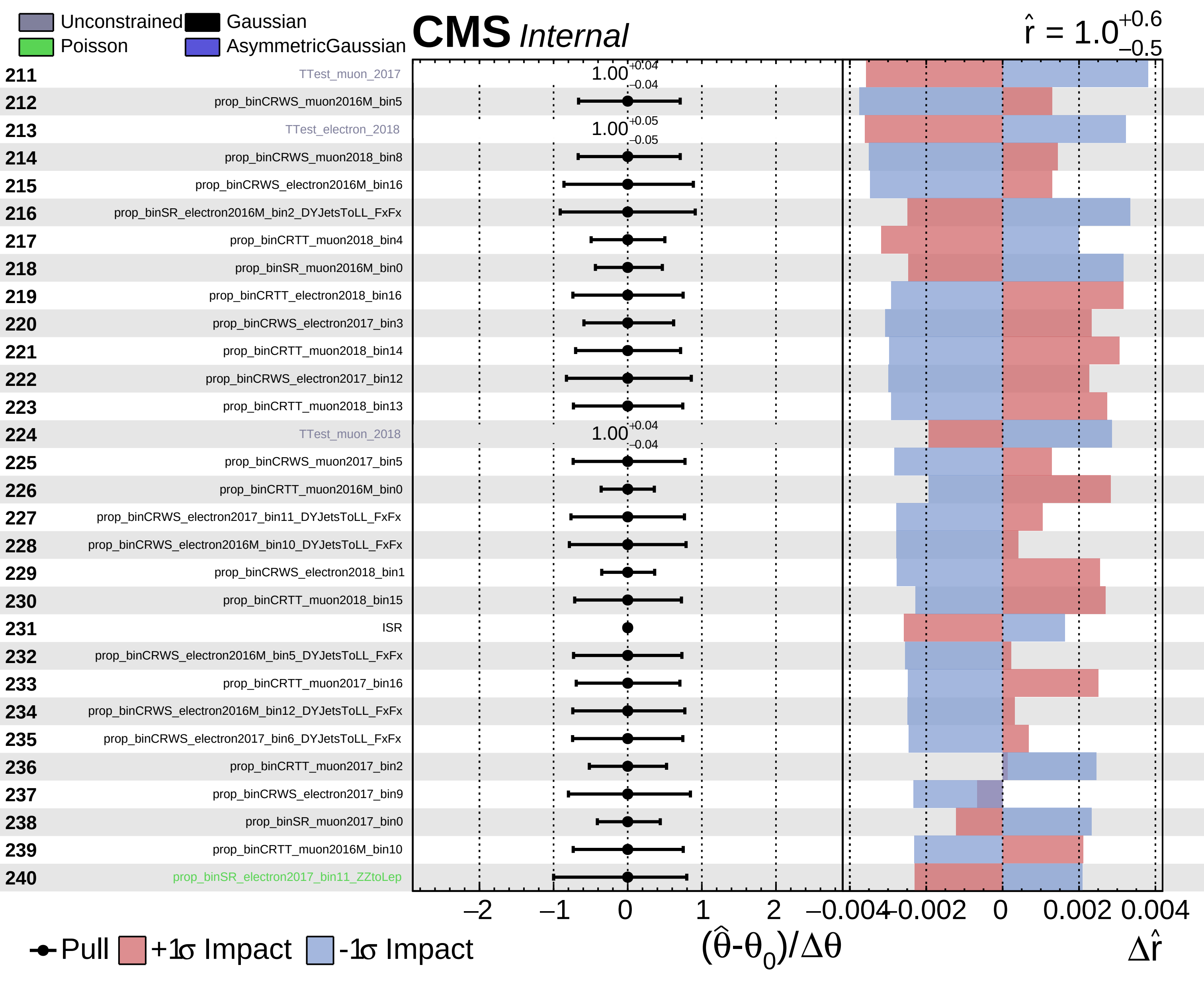

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
121 prop_binCRTT_electron2017_bin1
122 prop_binSR_electron2017_bin6
123 prop_binCRWS_electron2016M_bin3
124 jer_2016M
125 prop_binSR_electron2016M_bin11
126 VBS_2017
127 prop_binCRTT_electron2018_bin2
128 prop_binCRTT_electron2018_bin0
129 prop_binCRWS_muon2018_bin1
130 prop_binSR_electron2017_bin2
131 VBS_2018
132 prop_binSR_electron2016M_bin8
133 prop_binSR_electron2018_bin2
134 prop_binCRWS_muon2018_bin9
135 QCDScale_Triboson
136 prop_binSR_electron2017_bin5
137 prop_binCRWS_muon2018_bin16
138 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
139 QCDScale_TVX
140 prop_binCRWS_electron2018_bin10
141 prop_binCRWS_muon2016M_bin3
142 prop_binSR_electron2016M_bin7
143 prop_binCRWS_electron2016M_bin1
144 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
145 prop_binCRWS_muon2018_bin15
146 prop_binCRWS_muon2017_bin10
147 prop_binSR_electron2017_bin11_Fake
148 prop_binCRWS_muon2018_bin13
149 prop_binSR_electron2016M_bin4
150 prop_binCRWS_muon2017_bin6
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



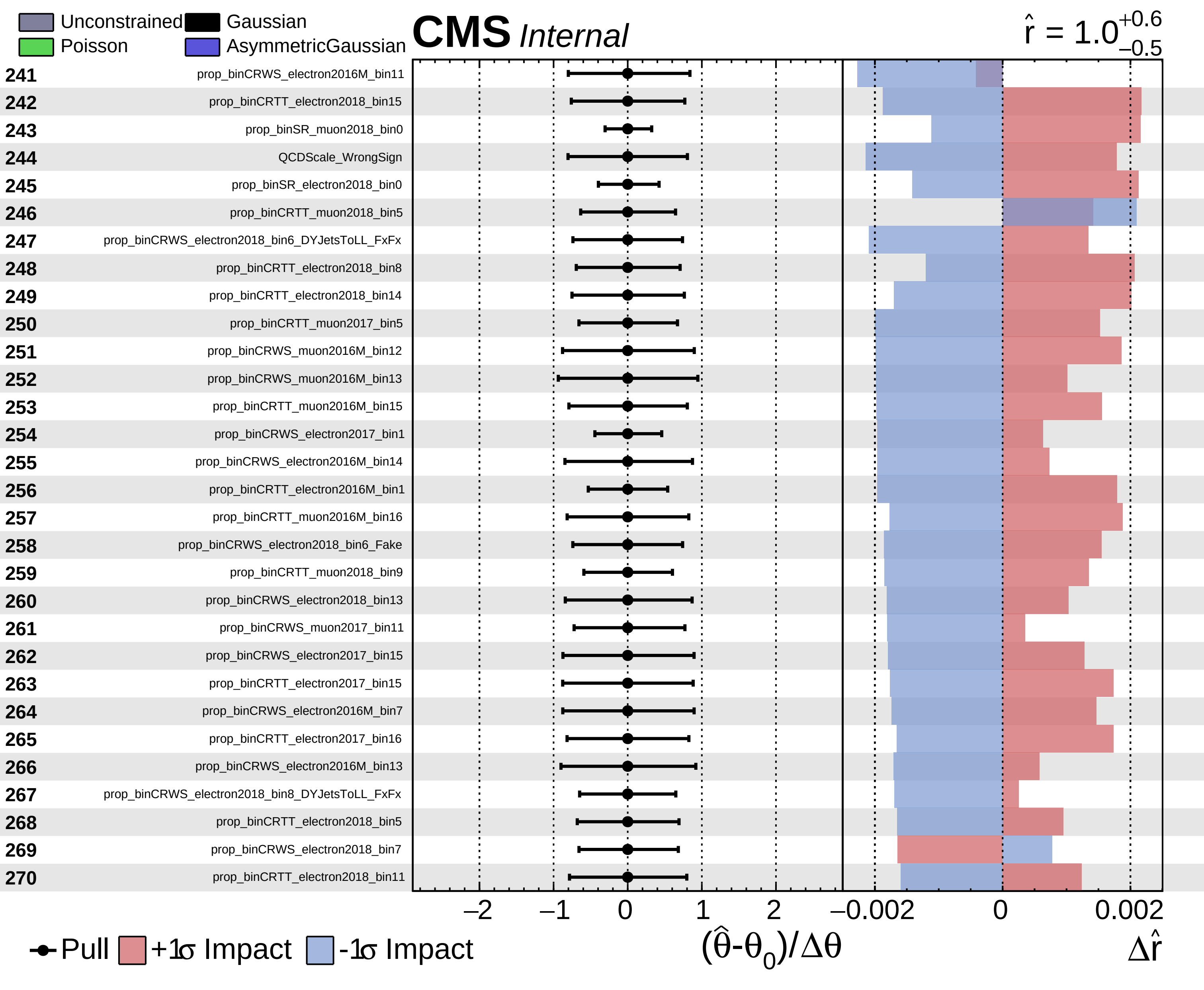
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
151 QCDScale_VG
152 prop_binCRTT_electron2017_bin0
153 prop_binCRWS_electron2016M_bin2
154 prop_binSR_muon2016M_bin1
155 prop_binSR_electron2016M_bin6
156 prop_binSR_muon2017_bin12
157 prop_binCRWS_electron2016M_bin4
158 prop_binCRWS_electron2018_bin16
159 prop_binSR_muon2017_bin6_Fake
160 prop_binCRWS_muon2017_bin9
161 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
162 tau_vsjet_2017
163 prop_binCRTT_muon2018_bin11
164 prop_binSR_electron2016M_bin5
165 prop_binCRWS_muon2016M_bin6
166 prop_binCRTT_electron2018_bin3
167 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
168 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
169 jes
170 prop_binCRTT_muon2017_bin1
171 prop_binCRWS_muon2018_bin14
172 prop_binSR_electron2016M_bin2_Fake
173 prop_binCRWS_electron2017_bin8
174 prop_binCRWS_electron2016M_bin6
175 prop_binSR_electron2017_bin11_Triboson
176 prop_binCRWS_muon2016M_bin16
177 prop_binCRTT_electron2017_bin4
178 tau_vsjet_2018
179 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
180 prop_binSR_electron2016M_bin1
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



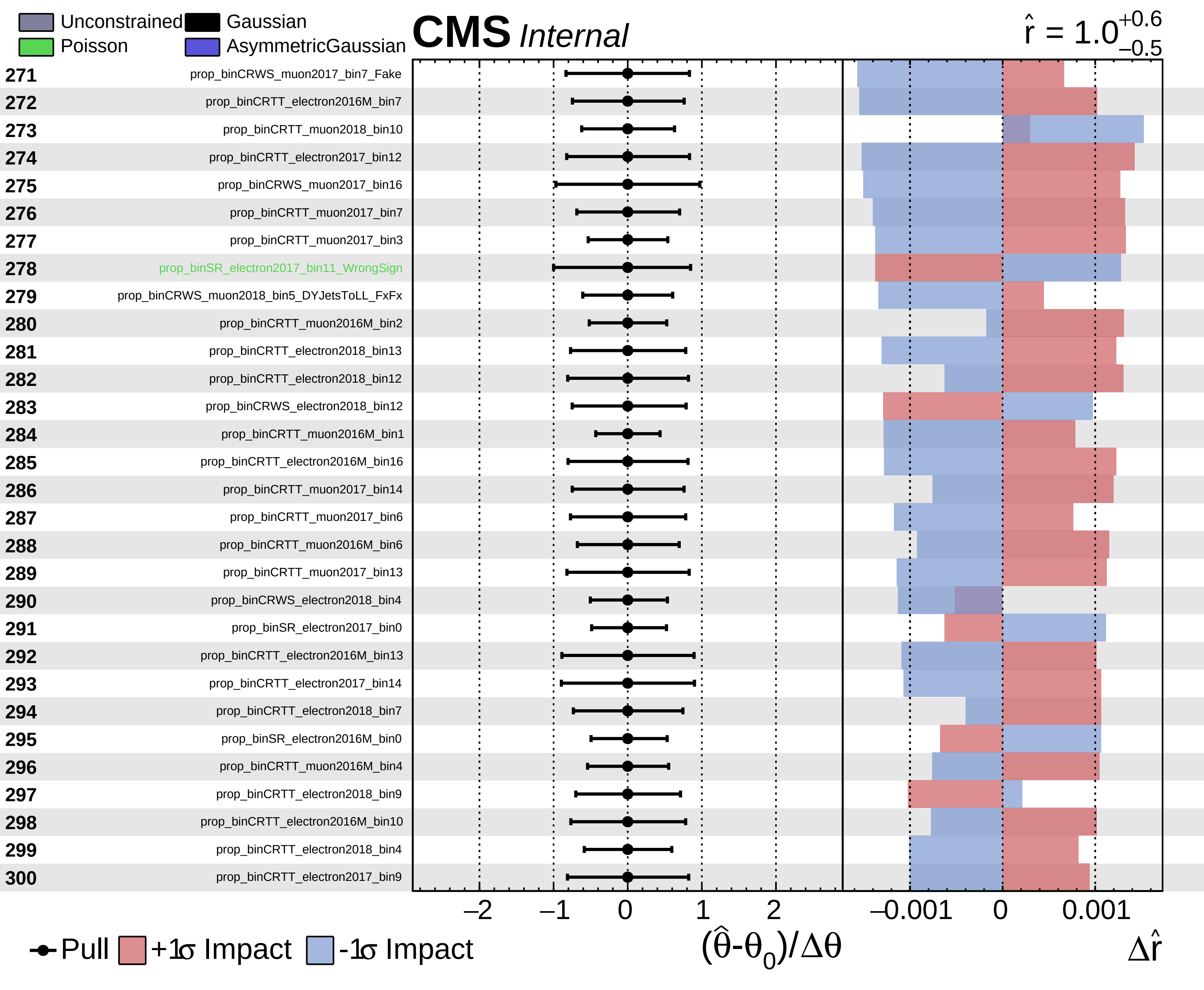
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
181 VBS_2016M
182 prop_binCRTT_muon2017_bin0
183 jer_2017
184 prop_binCRWS_electron2018_bin15
185 FSR
186 prop_binSR_electron2017_bin1
187 prop_binSR_electron2018_bin1
188 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
189 TTest_electron_2017
1.00 +0.05 −0.05
190 prop_binCRWS_muon2017_bin8
191 prop_binCRTT_muon2018_bin1
192 prop_binCRWS_muon2016M_bin8
193 tau_vsjet_2016M
194 prop_binCRWS_muon2018_bin4
195 prop_binCRTT_muon2017_bin4
196 prop_binCRWS_muon2018_bin0
197 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
198 prop_binSR_electron2016M_bin3_Fake
199 prop_binCRWS_muon2017_bin14
200 TES
201 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
202 prop_binCRWS_muon2016M_bin0
203 prop_binCRWS_electron2017_bin16
204 prop_binCRWS_muon2016M_bin11
205 prop_binCRTT_muon2018_bin2
206 prop_binCRWS_electron2017_bin7
207 prop_binCRTT_muon2018_bin16
208 prop_binCRWS_muon2018_bin11
209 prop_binCRTT_electron2018_bin1
210 prop_binCRWS_electron2017_bin5
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

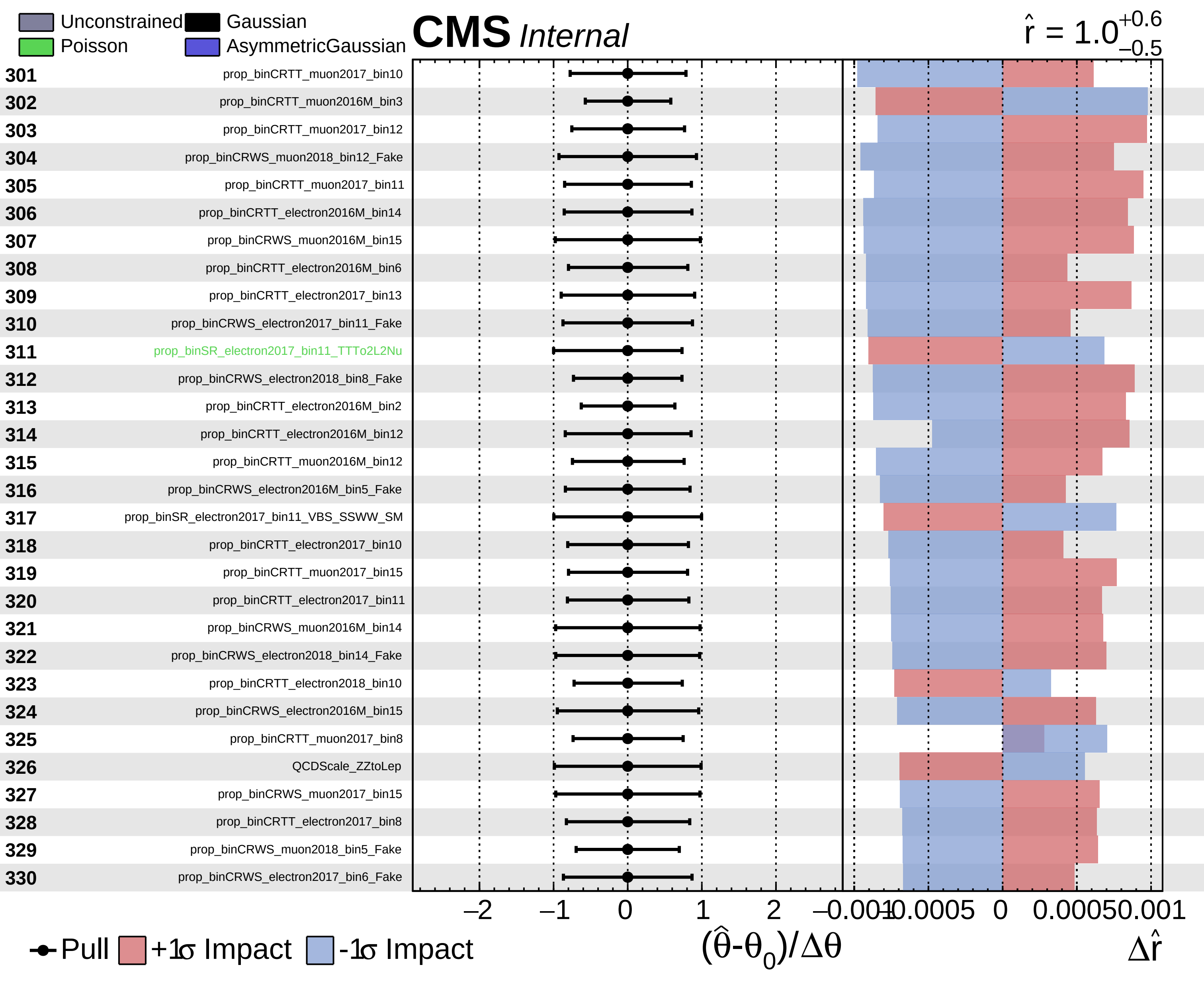


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
211 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
212 prop_binCRWS_muon2016M_bin5
213 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
214 prop_binCRWS_muon2018_bin8
215 prop_binCRWS_electron2016M_bin16
216 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
217 prop_binCRTT_muon2018_bin4
218 prop_binSR_muon2016M_bin0
219 prop_binCRTT_electron2018_bin16
220 prop_binCRWS_electron2017_bin3
221 prop_binCRTT_muon2018_bin14
222 prop_binCRWS_electron2017_bin12
223 prop_binCRTT_muon2018_bin13
224 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
225 prop_binCRWS_muon2017_bin5
226 prop_binCRTT_muon2016M_bin0
227 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
228 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
229 prop_binCRWS_electron2018_bin1
230 prop_binCRTT_muon2018_bin15
231 ISR
232 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
233 prop_binCRTT_muon2017_bin16
234 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
235 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
236 prop_binCRTT_muon2017_bin2
237 prop_binCRWS_electron2017_bin9
238 prop_binSR_muon2017_bin0
239 prop_binCRTT_muon2016M_bin10
240 prop_binSR_electron2017_bin11_ZZtoLep
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

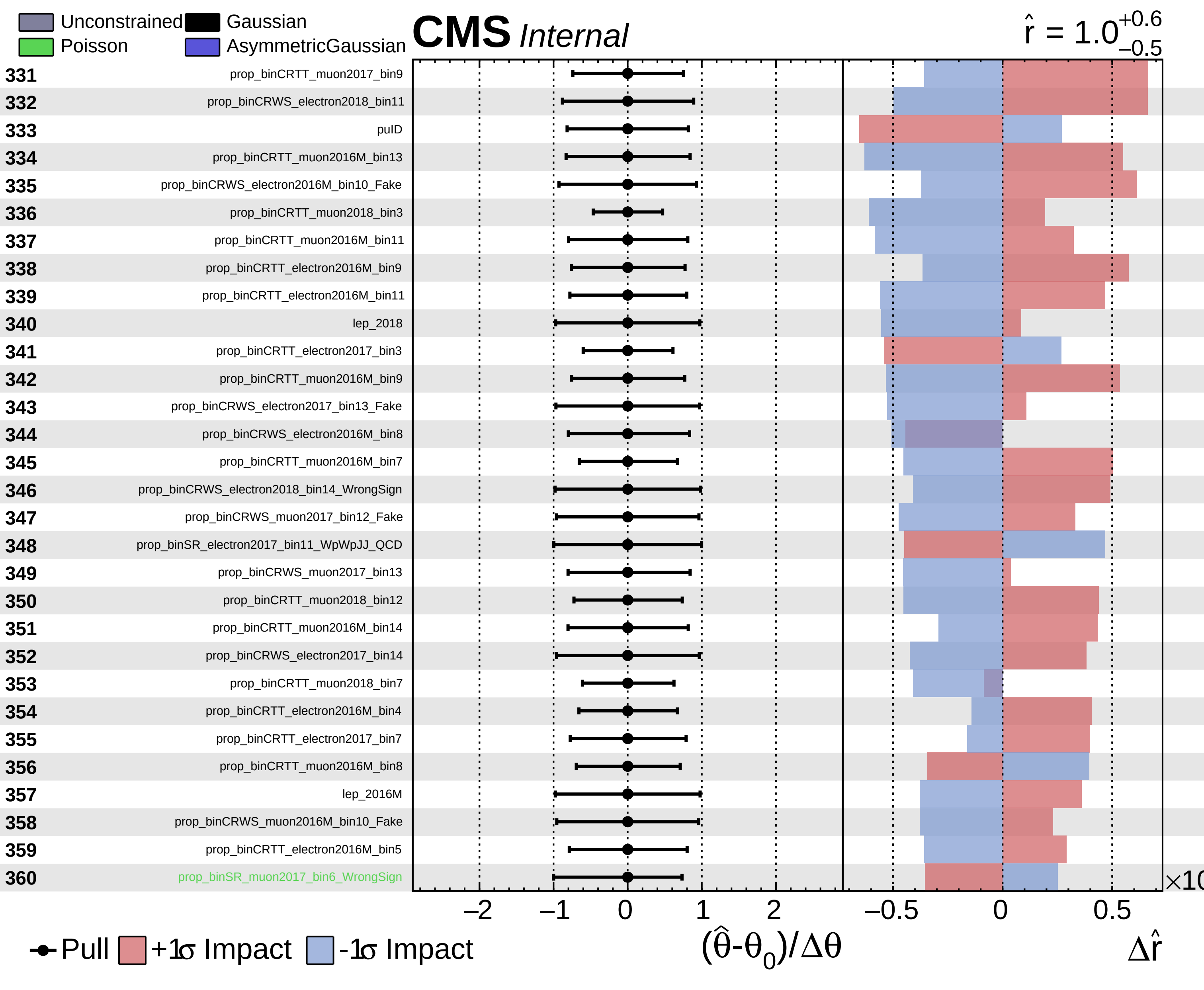


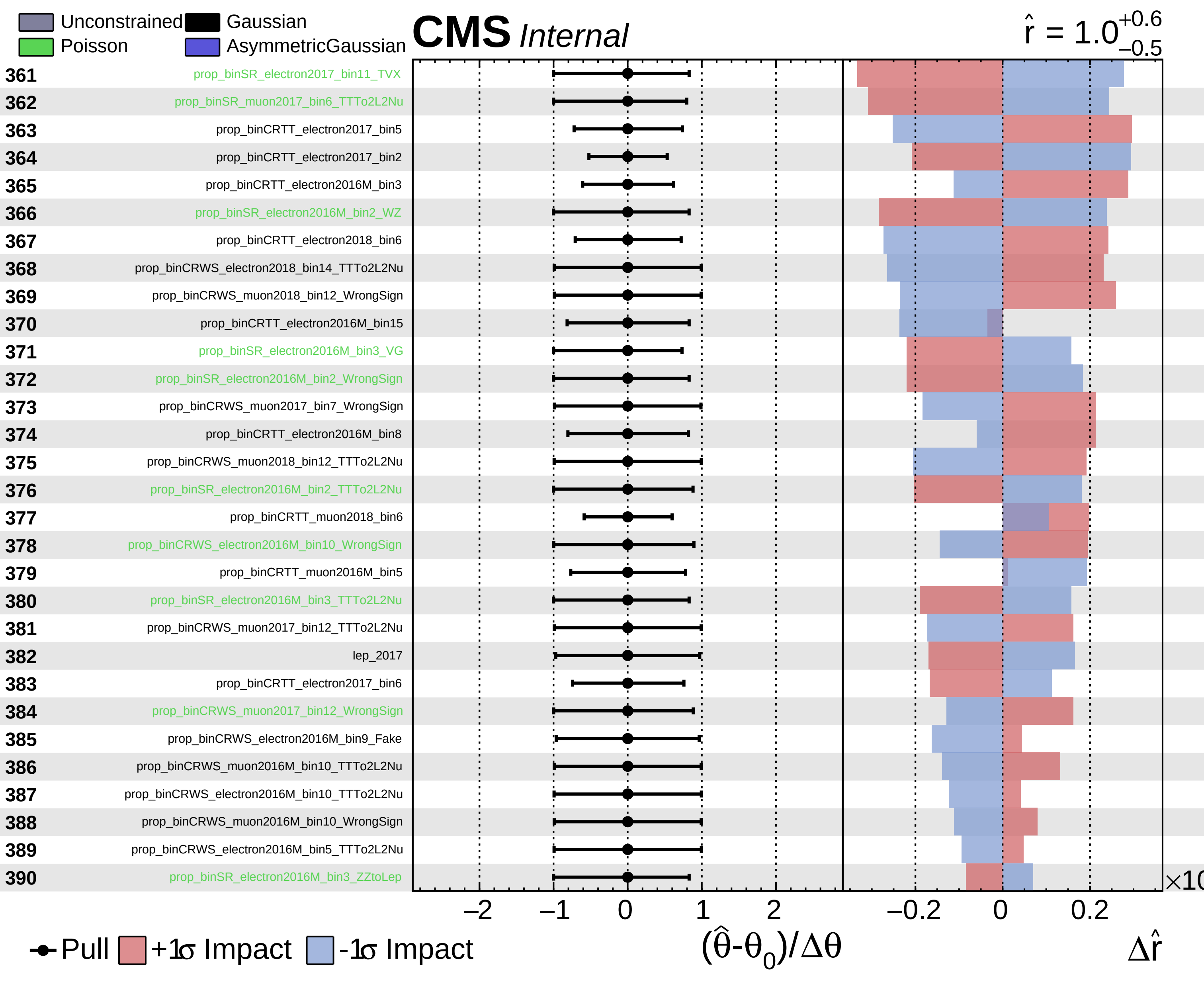
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
241 prop_binCRWS_electron2016M_bin11
242 prop_binCRTT_electron2018_bin15
243 prop_binSR_muon2018_bin0
244 QCDScale_WrongSign
245 prop_binSR_electron2018_bin0
246 prop_binCRTT_muon2018_bin5
247 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
248 prop_binCRTT_electron2018_bin8
249 prop_binCRTT_electron2018_bin14
250 prop_binCRTT_muon2017_bin5
251 prop_binCRWS_muon2016M_bin12
252 prop_binCRWS_muon2016M_bin13
253 prop_binCRTT_muon2016M_bin15
254 prop_binCRWS_electron2017_bin1
255 prop_binCRWS_electron2016M_bin14
256 prop_binCRTT_electron2016M_bin1
257 prop_binCRTT_muon2016M_bin16
258 prop_binCRWS_electron2018_bin6_Fake
259 prop_binCRTT_muon2018_bin9
260 prop_binCRWS_electron2018_bin13
261 prop_binCRWS_muon2017_bin11
262 prop_binCRWS_electron2017_bin15
263 prop_binCRTT_electron2017_bin15
264 prop_binCRWS_electron2016M_bin7
265 prop_binCRTT_electron2017_bin16
266 prop_binCRWS_electron2016M_bin13
267 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
268 prop_binCRTT_electron2018_bin5
269 prop_binCRWS_electron2018_bin7
270 prop_binCRTT_electron2018_bin11
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
271 prop_binCRWS_muon2017_bin7_Fake
272 prop_binCRTT_electron2016M_bin7
273 prop_binCRTT_muon2018_bin10
274 prop_binCRTT_electron2017_bin12
275 prop_binCRWS_muon2017_bin16
276 prop_binCRTT_muon2017_bin7
277 prop_binCRTT_muon2017_bin3
278 prop_binSR_electron2017_bin11_WrongSign
279 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
280 prop_binCRTT_muon2016M_bin2
281 prop_binCRTT_electron2018_bin13
282 prop_binCRTT_electron2018_bin12
283 prop_binCRWS_electron2018_bin12
284 prop_binCRTT_muon2016M_bin1
285 prop_binCRTT_electron2016M_bin16
286 prop_binCRTT_muon2017_bin14
287 prop_binCRTT_muon2017_bin6
288 prop_binCRTT_muon2016M_bin6
289 prop_binCRTT_muon2017_bin13
290 prop_binCRWS_electron2018_bin4
291 prop_binSR_electron2017_bin0
292 prop_binCRTT_electron2016M_bin13
293 prop_binCRTT_electron2017_bin14
294 prop_binCRTT_electron2018_bin7
295 prop_binSR_electron2016M_bin0
296 prop_binCRTT_muon2016M_bin4
297 prop_binCRTT_electron2018_bin9
298 prop_binCRTT_electron2016M_bin10
299 prop_binCRTT_electron2018_bin4
300 prop_binCRTT_electron2017_bin9
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

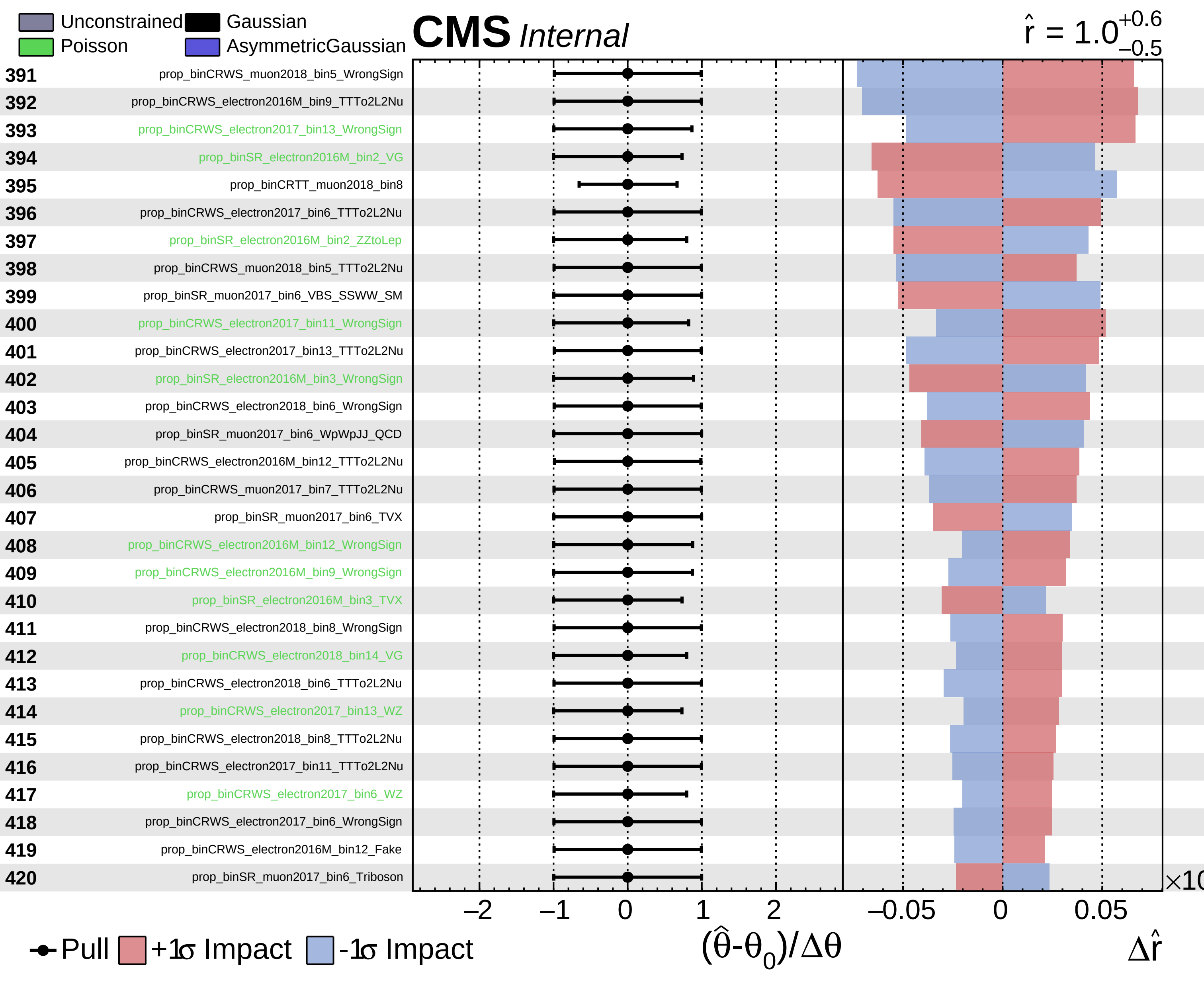


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
301 prop_binCRTT_muon2017_bin10
302 prop_binCRTT_muon2016M_bin3
303 prop_binCRTT_muon2017_bin12
304 prop_binCRWS_muon2018_bin12_Fake
305 prop_binCRTT_muon2017_bin11
306 prop_binCRTT_electron2016M_bin14
307 prop_binCRWS_muon2016M_bin15
308 prop_binCRTT_electron2016M_bin6
309 prop_binCRTT_electron2017_bin13
310 prop_binCRWS_electron2017_bin11_Fake
311 prop_binSR_electron2017_bin11_TTTo2L2Nu
312 prop_binCRWS_electron2018_bin8_Fake
313 prop_binCRTT_electron2016M_bin2
314 prop_binCRTT_electron2016M_bin12
315 prop_binCRTT_muon2016M_bin12
316 prop_binCRWS_electron2016M_bin5_Fake
317 prop_binSR_electron2017_bin11_VBS_SSWW_SM
318 prop_binCRTT_electron2017_bin10
319 prop_binCRTT_muon2017_bin15
320 prop_binCRTT_electron2017_bin11
321 prop_binCRWS_muon2016M_bin14
322 prop_binCRWS_electron2018_bin14_Fake
323 prop_binCRTT_electron2018_bin10
324 prop_binCRWS_electron2016M_bin15
325 prop_binCRTT_muon2017_bin8
326 QCDScale_ZZtoLep
327 prop_binCRWS_muon2017_bin15
328 prop_binCRTT_electron2017_bin8
329 prop_binCRWS_muon2018_bin5_Fake
330 prop_binCRWS_electron2017_bin6_Fake
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

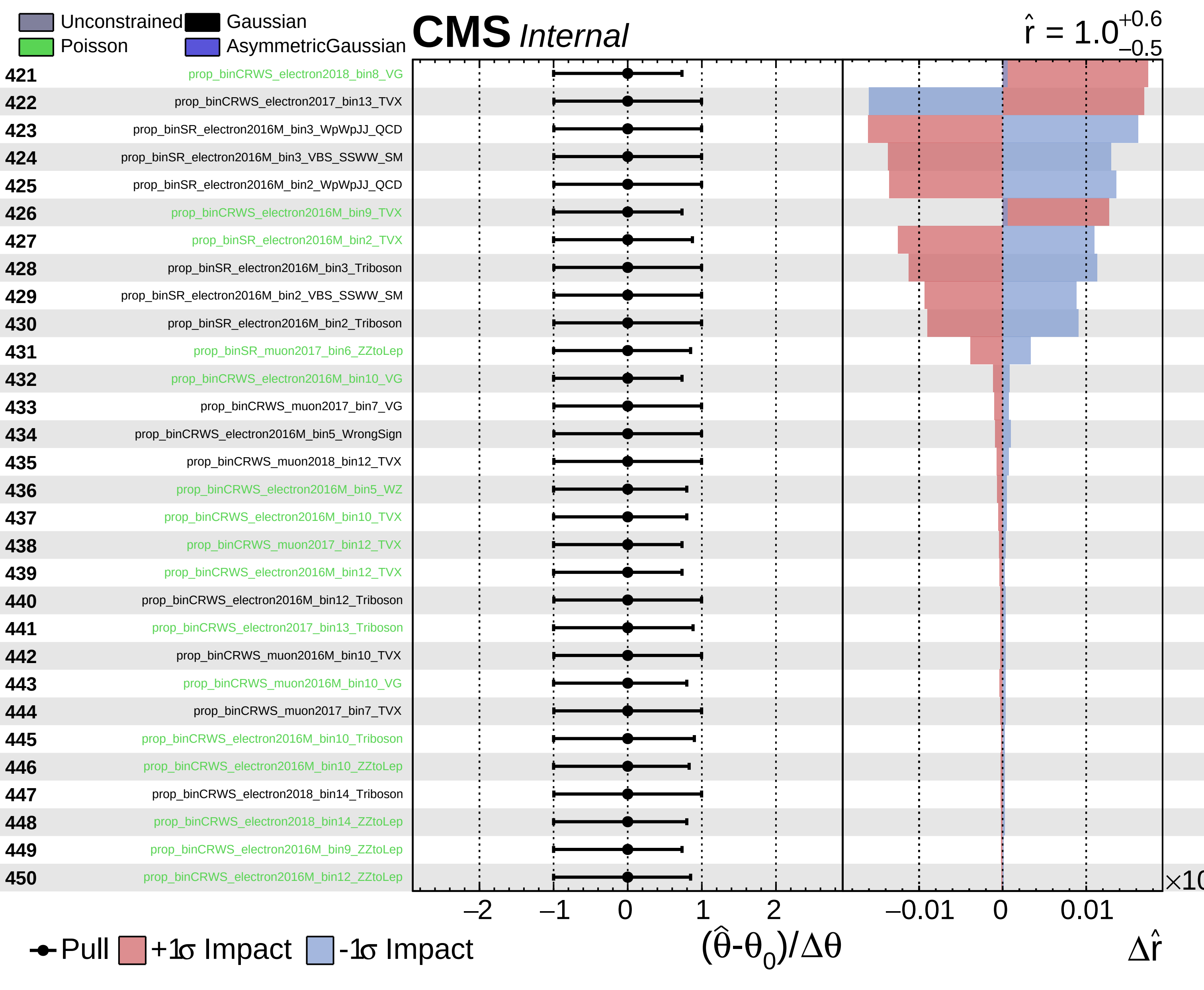

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
331 prop_binCRTT_muon2017_bin9
332 prop_binCRWS_electron2018_bin11
333 puID
334 prop_binCRTT_muon2016M_bin13
335 prop_binCRWS_electron2016M_bin10_Fake
336 prop_binCRTT_muon2018_bin3
337 prop_binCRTT_muon2016M_bin11
338 prop_binCRTT_electron2016M_bin9
339 prop_binCRTT_electron2016M_bin11
340 lep_2018
341 prop_binCRTT_electron2017_bin3
342 prop_binCRTT_muon2016M_bin9
343 prop_binCRWS_electron2017_bin13_Fake
344 prop_binCRWS_electron2016M_bin8
345 prop_binCRTT_muon2016M_bin7
346 prop_binCRWS_electron2018_bin14_WrongSign
347 prop_binCRWS_muon2017_bin12_Fake
348 prop_binSR_electron2017_bin11_WpWpJJ_QCD
349 prop_binCRWS_muon2017_bin13
350 prop_binCRTT_muon2018_bin12
351 prop_binCRTT_muon2016M_bin14
352 prop_binCRWS_electron2017_bin14
353 prop_binCRTT_muon2018_bin7
354 prop_binCRTT_electron2016M_bin4
355 prop_binCRTT_electron2017_bin7
356 prop_binCRTT_muon2016M_bin8
357 lep_2016M
358 prop_binCRWS_muon2016M_bin10_Fake
359 prop_binCRTT_electron2016M_bin5
360 prop_binSR_muon2017_bin6_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

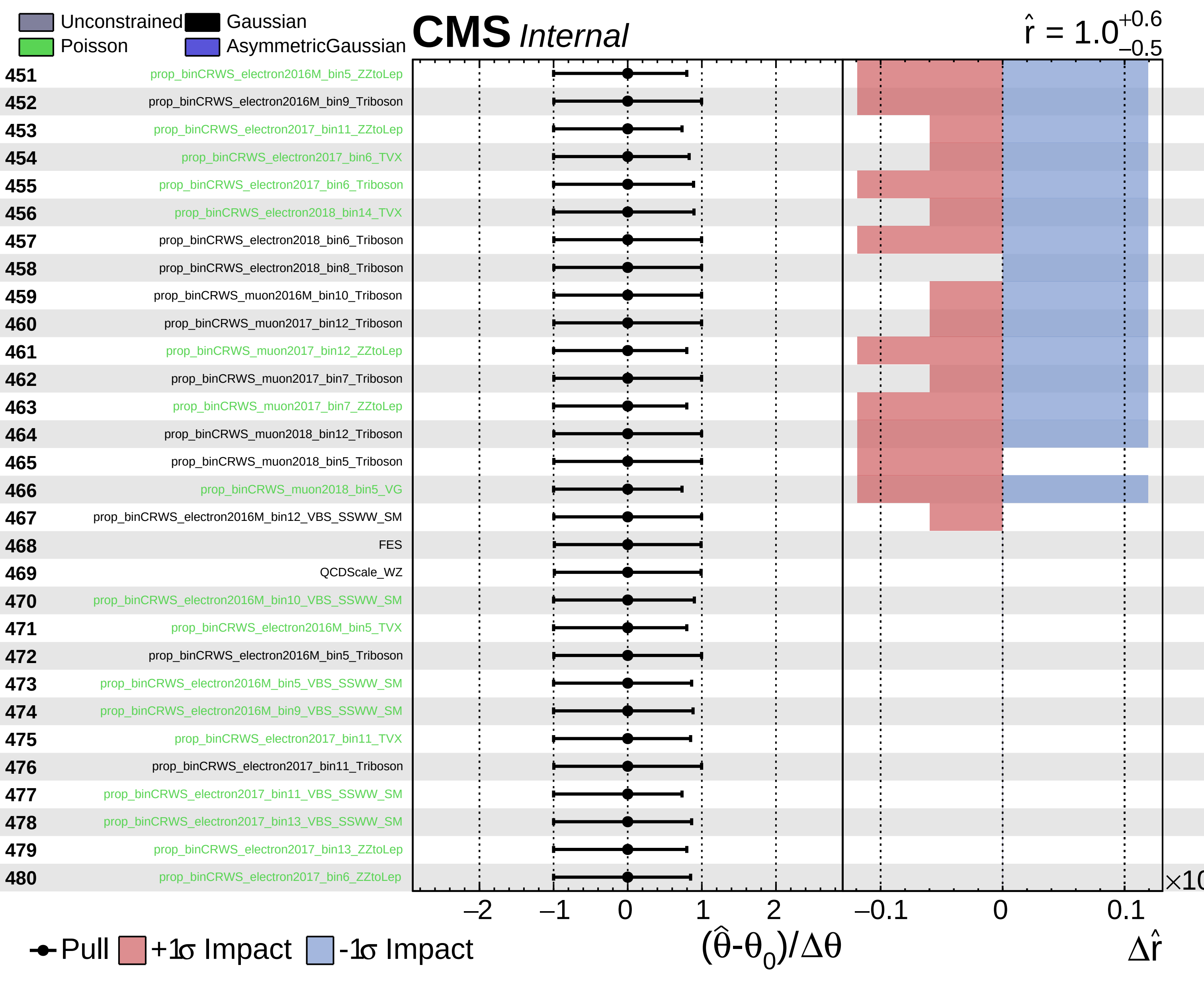


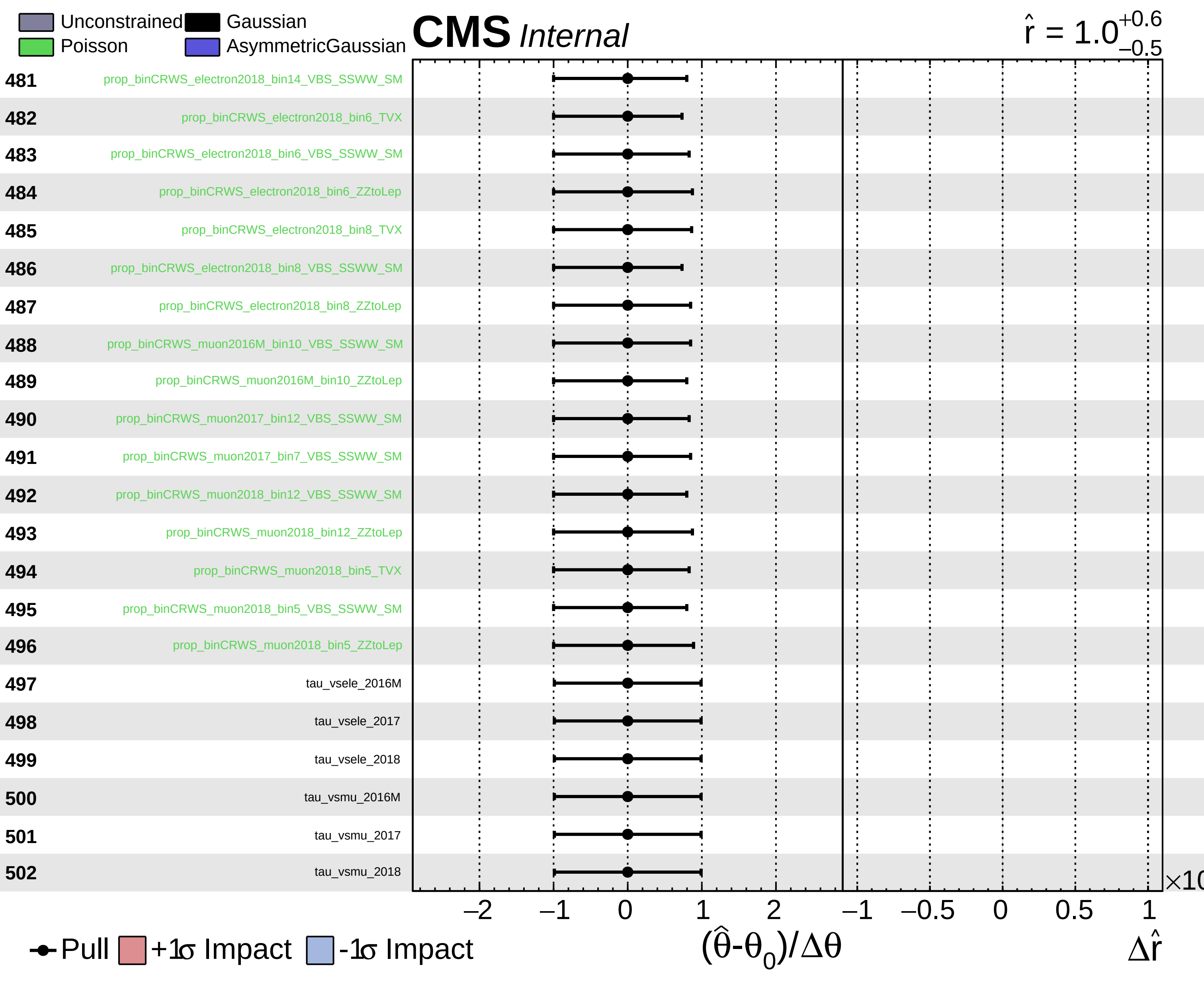
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
361 prop_binSR_electron2017_bin11_TVX
362 prop_binSR_muon2017_bin6_TTTo2L2Nu
363 prop_binCRTT_electron2017_bin5
364 prop_binCRTT_electron2017_bin2
365 prop_binCRTT_electron2016M_bin3
366 prop_binSR_electron2016M_bin2_WZ
367 prop_binCRTT_electron2018_bin6
368 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
369 prop_binCRWS_muon2018_bin12_WrongSign
370 prop_binCRTT_electron2016M_bin15
371 prop_binSR_electron2016M_bin3_VG
372 prop_binSR_electron2016M_bin2_WrongSign
373 prop_binCRWS_muon2017_bin7_WrongSign
374 prop_binCRTT_electron2016M_bin8
375 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
376 prop_binSR_electron2016M_bin2_TTTo2L2Nu
377 prop_binCRTT_muon2018_bin6
378 prop_binCRWS_electron2016M_bin10_WrongSign
379 prop_binCRTT_muon2016M_bin5
380 prop_binSR_electron2016M_bin3_TTTo2L2Nu
381 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
382 lep_2017
383 prop_binCRTT_electron2017_bin6
384 prop_binCRWS_muon2017_bin12_WrongSign
385 prop_binCRWS_electron2016M_bin9_Fake
386 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
387 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
388 prop_binCRWS_muon2016M_bin10_WrongSign
389 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
390 prop_binSR_electron2016M_bin3_ZZtoLep
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
391 prop_binCRWS_muon2018_bin5_WrongSign
392 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
393 prop_binCRWS_electron2017_bin13_WrongSign
394 prop_binSR_electron2016M_bin2_VG
395 prop_binCRTT_muon2018_bin8
396 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
397 prop_binSR_electron2016M_bin2_ZZtoLep
398 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
399 prop_binSR_muon2017_bin6_VBS_SSWW_SM
400 prop_binCRWS_electron2017_bin11_WrongSign
401 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
402 prop_binSR_electron2016M_bin3_WrongSign
403 prop_binCRWS_electron2018_bin6_WrongSign
404 prop_binSR_muon2017_bin6_WpWpJJ_QCD
405 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
406 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
407 prop_binSR_muon2017_bin6_TVX
408 prop_binCRWS_electron2016M_bin12_WrongSign
409 prop_binCRWS_electron2016M_bin9_WrongSign
410 prop_binSR_electron2016M_bin3_TVX
411 prop_binCRWS_electron2018_bin8_WrongSign
412 prop_binCRWS_electron2018_bin14_VG
413 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
414 prop_binCRWS_electron2017_bin13_WZ
415 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
416 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
417 prop_binCRWS_electron2017_bin6_WZ
418 prop_binCRWS_electron2017_bin6_WrongSign
419 prop_binCRWS_electron2016M_bin12_Fake
420 prop_binSR_muon2017_bin6_Triboson
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
421 prop_binCRWS_electron2018_bin8_VG
422 prop_binCRWS_electron2017_bin13_TVX
423 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
424 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
425 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
426 prop_binCRWS_electron2016M_bin9_TVX
427 prop_binSR_electron2016M_bin2_TVX
428 prop_binSR_electron2016M_bin3_Triboson
429 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
430 prop_binSR_electron2016M_bin2_Triboson
431 prop_binSR_muon2017_bin6_ZZtoLep
432 prop_binCRWS_electron2016M_bin10_VG
433 prop_binCRWS_muon2017_bin7_VG
434 prop_binCRWS_electron2016M_bin5_WrongSign
435 prop_binCRWS_muon2018_bin12_TVX
436 prop_binCRWS_electron2016M_bin5_WZ
437 prop_binCRWS_electron2016M_bin10_TVX
438 prop_binCRWS_muon2017_bin12_TVX
439 prop_binCRWS_electron2016M_bin12_TVX
440 prop_binCRWS_electron2016M_bin12_Triboson
441 prop_binCRWS_electron2017_bin13_Triboson
442 prop_binCRWS_muon2016M_bin10_TVX
443 prop_binCRWS_muon2016M_bin10_VG
444 prop_binCRWS_muon2017_bin7_TVX
445 prop_binCRWS_electron2016M_bin10_Triboson
446 prop_binCRWS_electron2016M_bin10_ZZtoLep
447 prop_binCRWS_electron2018_bin14_Triboson
448 prop_binCRWS_electron2018_bin14_ZZtoLep
449 prop_binCRWS_electron2016M_bin9_ZZtoLep
450 prop_binCRWS_electron2016M_bin12_ZZtoLep
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
451 prop_binCRWS_electron2016M_bin5_ZZtoLep
452 prop_binCRWS_electron2016M_bin9_Triboson
453 prop_binCRWS_electron2017_bin11_ZZtoLep
454 prop_binCRWS_electron2017_bin6_TVX
455 prop_binCRWS_electron2017_bin6_Triboson
456 prop_binCRWS_electron2018_bin14_TVX
457 prop_binCRWS_electron2018_bin6_Triboson
458 prop_binCRWS_electron2018_bin8_Triboson
459 prop_binCRWS_muon2016M_bin10_Triboson
460 prop_binCRWS_muon2017_bin12_Triboson
461 prop_binCRWS_muon2017_bin12_ZZtoLep
462 prop_binCRWS_muon2017_bin7_Triboson
463 prop_binCRWS_muon2017_bin7_ZZtoLep
464 prop_binCRWS_muon2018_bin12_Triboson
465 prop_binCRWS_muon2018_bin5_Triboson
466 prop_binCRWS_muon2018_bin5_VG
467 prop_binCRWS_electron2016M_bin12_VBS_SSWW_SM
468 FES
469 QCDScale_WZ
470 prop_binCRWS_electron2016M_bin10_VBS_SSWW_SM
471 prop_binCRWS_electron2016M_bin5_TVX
472 prop_binCRWS_electron2016M_bin5_Triboson
473 prop_binCRWS_electron2016M_bin5_VBS_SSWW_SM
474 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
475 prop_binCRWS_electron2017_bin11_TVX
476 prop_binCRWS_electron2017_bin11_Triboson
477 prop_binCRWS_electron2017_bin11_VBS_SSWW_SM
478 prop_binCRWS_electron2017_bin13_VBS_SSWW_SM
479 prop_binCRWS_electron2017_bin13_ZZtoLep
480 prop_binCRWS_electron2017_bin6_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
481 prop_binCRWS_electron2018_bin14_VBS_SSWW_SM
482 prop_binCRWS_electron2018_bin6_TVX
483 prop_binCRWS_electron2018_bin6_VBS_SSWW_SM
484 prop_binCRWS_electron2018_bin6_ZZtoLep
485 prop_binCRWS_electron2018_bin8_TVX
486 prop_binCRWS_electron2018_bin8_VBS_SSWW_SM
487 prop_binCRWS_electron2018_bin8_ZZtoLep
488 prop_binCRWS_muon2016M_bin10_VBS_SSWW_SM
489 prop_binCRWS_muon2016M_bin10_ZZtoLep
490 prop_binCRWS_muon2017_bin12_VBS_SSWW_SM
491 prop_binCRWS_muon2017_bin7_VBS_SSWW_SM
492 prop_binCRWS_muon2018_bin12_VBS_SSWW_SM
493 prop_binCRWS_muon2018_bin12_ZZtoLep
494 prop_binCRWS_muon2018_bin5_TVX
495 prop_binCRWS_muon2018_bin5_VBS_SSWW_SM
496 prop_binCRWS_muon2018_bin5_ZZtoLep
497 tau_vsele_2016M
498 tau_vsele_2017
499 tau_vsele_2018
500 tau_vsmu_2016M
501 tau_vsmu_2017
502 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂